Project data

Project name	CON1
Project number	
Author	EccentricBolt GUI
Description	Crawford & Kulak (1971) type specimen - 2x4 bolts, e = 12 in
Date	10/16/2026
Code	AISC/ACI

Material

Steel	A572 Gr.50

Project item SupportPlate - LoadStub

Name	SupportPlate - LoadStub
Description	
Analysis	Stress, strain/ loads in equilibrium
Design code	AISC 360-16 (LRFD) / ACI 318-14

Design

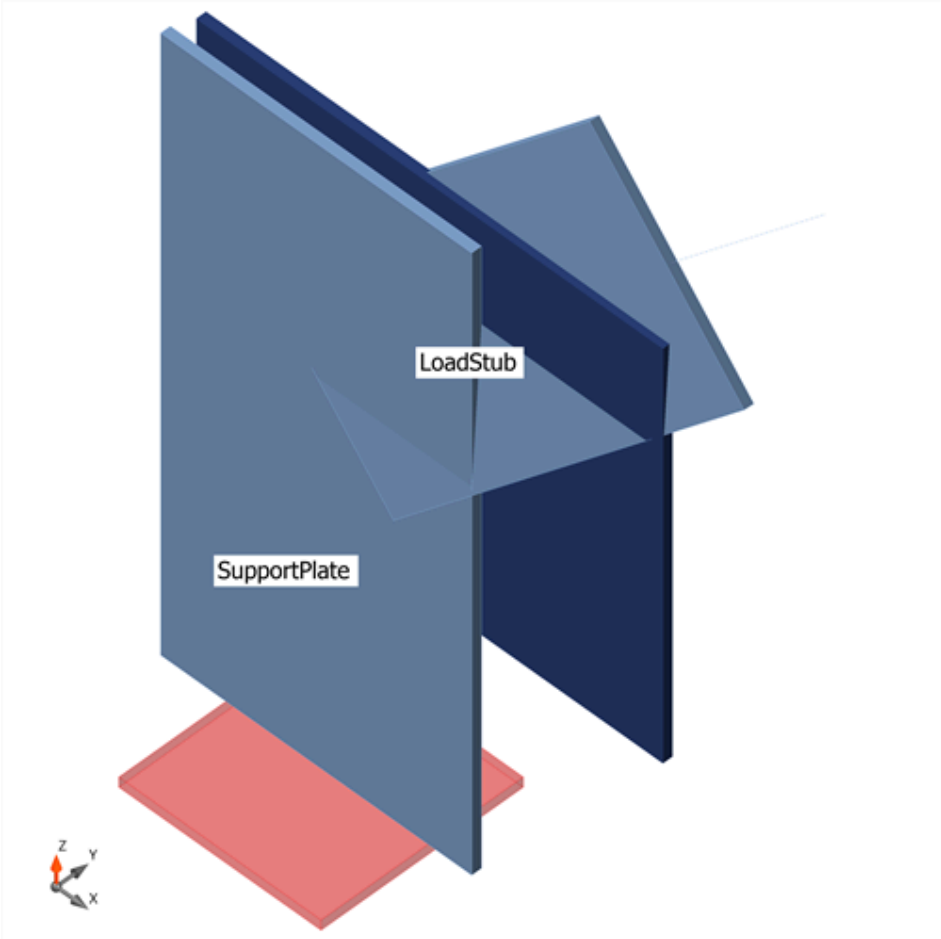
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SupportPlate	1 - PL-A 305x10	0.0	90.0	-180.0	229	0	0
LoadStub	2 - PL-C 229x10	3.6	44.9	174.9	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SupportPlate / begin	N-Vy-Vz-Mx-My-Mz	Position	0
LoadStub / end		Position	0



SupportPlate - LoadStub, Axonometric view

Cross-sections

Name	Material
1 - PL-A 305x10	A572 Gr.50
2 - PL-C 229x10	A572 Gr.50

Load effects (forces in equilibrium)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
LC1 - eccentric load	SupportPlate / End	266.9	0.0	0.0	0.0	81.3	0.0
	LoadStub / End	-266.9	0.0	0.0	0.0	-81.3	0.0

Unbalanced forces

Name	X [kN]	Y [kN]	Z [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
LC1 - eccentric load	-188.8	-11.8	78.6	0.0	-0.2	-5.1

Code settings

Item	Value	Unit	Reference
Friction coefficient - concrete	0.40	-	ACI 349-01 – B.6.1.4
Friction coefficient in slip-resistance	0.30	-	AISC 360-16 – J3.8
Limit plastic strain	0.05	-	
Detailing	Yes		

Item	Value	Unit	Reference
Distance between bolts [d]	2.66	-	AISC 360-16 – J3.3
Concrete breakout resistance check	Both		
Base metal capacity check at weld fusion face	No		AISC 360-16 – J2-2
Deformation at bolt hole at service load is design consideration	Yes		AISC 360-16 – J3.10
Cracked concrete	Yes		ACI 318-14 – 17
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints